

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

MA-220(AU/CM/ME/FT/MI)-CBCS

B.E. IV Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Mathematics - III

Time : Three Hours

Maximum Marks : 60

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Find a Fourier series to represent $x - x^2$ from $x = -\pi$ to

$$x = \pi. \text{ Also deduce } \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$$

b) Expand $f(x) = x \sin x$ as a Fourier series in the range $0 < x < 2\pi$.

2. a) Find Fourier cosine transform of $f(x) = \frac{1}{1+x^2}$

b) Develop $\sin\left(\frac{\pi x}{l}\right)$ in half range cosine series in the range $0 < x < l$.

3. a) Find Laplace transform of $te^{-4t} \sin 3t$.

b) Find inverse Laplace transform of $\frac{5s+3}{(s-1)(s^2+2s+5)}$.

MA-220(AU/CM/ME/FT/MI)-CBCS

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[2]

4. a) Evaluate $\oint_c \frac{e^z}{(z+1)^2}$, where c is the circle $|z-1|=3$.
- b) Find the imaginary part of the analytic function whose real part is $x^3 - 3xy^2 + 3x^2 - 3y^2$.

5. a) If $u(x, y)$ and $v(x, y)$ are harmonic functions in a region R , prove that the function

$$\left[\left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) + i \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \right]$$

is an analytic function of $z = x + iy$.

- b) Apply the calculus of Residue to show that

$$\int_0^\pi \frac{(1 + 2 \cos \theta)}{(5 + 4 \cos \theta)} d\theta = 0$$

6. a) Use Picard's method to approximate the value of y when $x = 0.1$, given that $y = 1$ when $x = 0$ and $\frac{dy}{dx} = 3x + y^2$.

- b) Use Runge-Kutta method to solve the equation $\frac{dy}{dx} = 1 + y^2$ for $x = 0.2$ to $x = 0.4$ with $h = 0.2$. Given that initially at $x = 0$, $y = 0$.

[3]

7. a) Evaluate the followings:

$$\text{i) } L\left\{\frac{1-e^t}{t}\right\} \quad \text{ii) } L\left\{\int_0^t \frac{\sin t}{t} dt\right\}$$

b) Evaluate the followings:

$$\text{i) } L^{-1}\left\{\frac{1}{s^2-4s+5}\right\} \quad \text{ii) } L^{-1}\left\{\frac{s}{(s+2)s}\right\}$$

8. a) Find all the poles of $\frac{z^5}{(z-2)^2(z^2+4)^3(z-4)^4}$

b) Using residue theorem evaluate $\int_c \frac{z-3}{z^2+2z+5} dz$ where c is $|z|=1$.
